

QQ9 Acid Base

AQA 2019-2022

IGC HK Exam

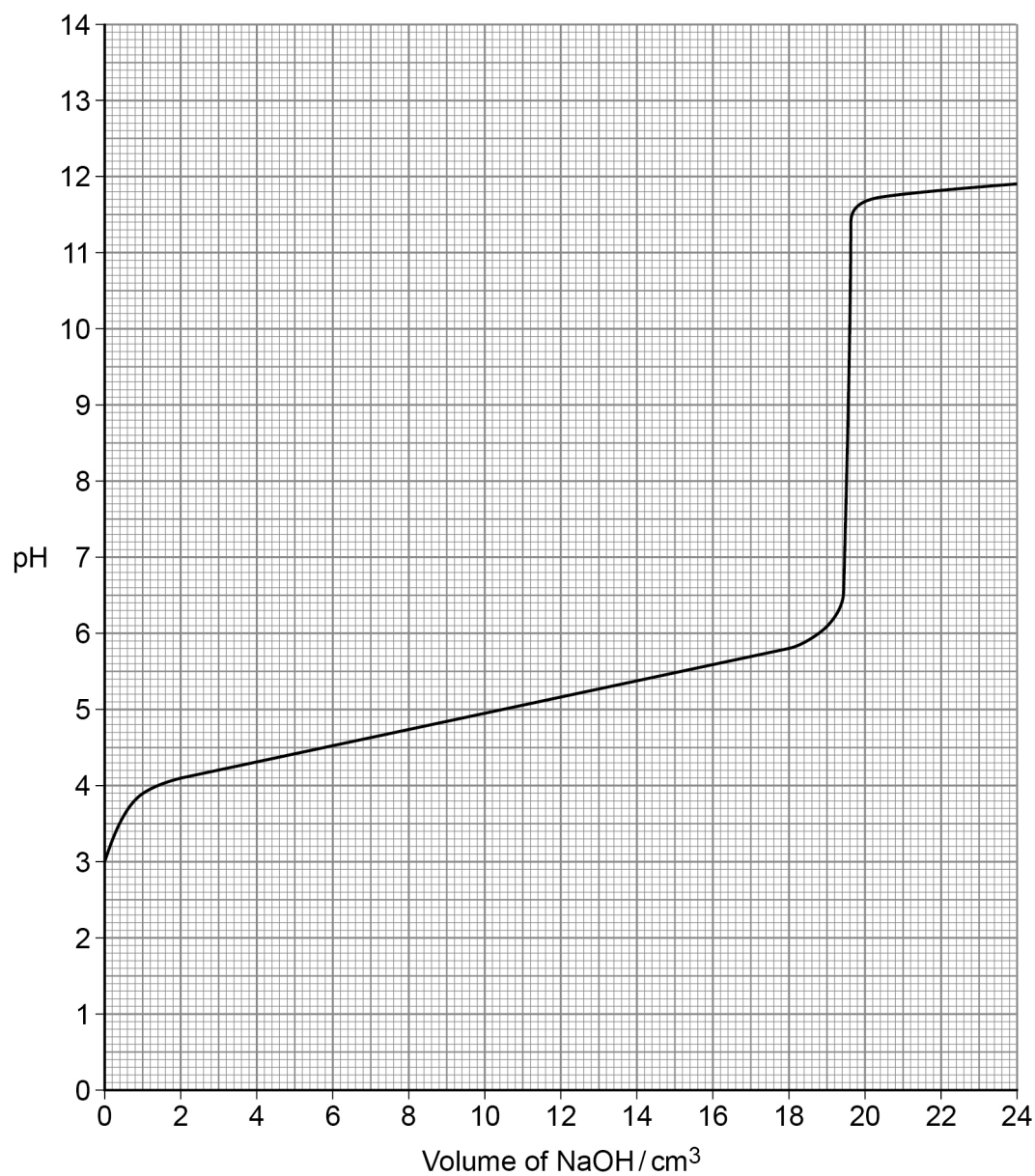


0 9 . 2

Sodium hydroxide solution was added gradually from a burette to 25 cm³ of 0.080 mol dm⁻³ propanoic acid at 25 °C
The pH was measured and recorded at regular intervals.

The results are shown in **Figure 4**.

Figure 4



Use **Figure 4** to determine the value of K_a for propanoic acid at 25 °C

Show your working.

[3 marks]

K_a _____ mol dm⁻³

0 9 . 3

Suggest which indicator is the most appropriate for the reaction in Question **09.2**?

Tick (✓) **one** box.

[1 mark]

Indicator	pH range	Tick (✓) one box
methyl orange	3.1 – 4.4	
bromothymol blue	6.0 – 7.6	
cresolphthalein	8.2 – 9.8	
indigo carmine	11.6 – 13.0	

Question 9 continues on the next page

Turn over ►



0	9	.	4
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A student prepared a buffer solution by adding 0.0136 mol of a salt KX to 100 cm³ of a 0.500 mol dm⁻³ solution of a weak acid HX and mixing thoroughly.

The student then added 3.00×10^{-4} mol of potassium hydroxide to the buffer solution.

Calculate the pH of the buffer solution after adding the potassium hydroxide.

For the weak acid HX at 25 °C the value of the acid dissociation constant, $K_a = 1.41 \times 10^{-5}$ mol dm⁻³.

Give your answer to two decimal places.

[6 marks]

pH _____



0	9	.	5
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A buffer solution has a constant pH even when diluted.

Use a mathematical expression to explain this.

[1 mark]

15

END OF QUESTIONS



Question	Answers	Additional Comments/Guidelines	Mark
09.1	M1: $[H^+] = [OH^-]$	M1: accept equal number/amounts of H^+ and OH^-	1
	M2: $[H^+] (= 10^{-pH}) = 2.138 \times 10^{-7}$	M2: allow 2.14×10^{-7}	1
	M3: $K_w = [H^+]^2$ or $(2.138 \times 10^{-7})^2$	M3: allow $(M2)^2$	1
	M4: $K_w = 4.57 \times 10^{-14}$	M4: allow 4.58×10^{-14} M4 is dependent on (an answer) ² in M3	1
09.2	View with Figure X (ie graph) as they may show working there.	Ignore calculations of mols of salt or acid	
	M1: Determines volume at half equivalence $(= \frac{19.5}{2} \text{cm}^3) = 9.75 \text{ (cm}^3\text{)}$	M1: Allow reading on graph to be from 19.4 to 19.7 giving M1 = 9.7 to 9.85	1
	M2: pH = 4.80 to 4.95	M2: Reads off pH at half equivalence	1
	M3: $K_a (= 10^{-pH}) = 10^{-4.9} = 1.26 \times 10^{-5}$	M3: Allow 1.12×10^{-5} to 1.58×10^{-5} M3: Allow 2sf or more	1
	Alternative method M1: pH of pure acid = 3 M2: $K_a = (10^{-3})^2 / 0.080$ M3: $= 1.25 \times 10^{-5}$	Alternative M1 if calculation incorrect: Allow pH = pK_a or $[H^+] = K_a$ at <u>half equivalence</u>	
09.3	cresolphthalein		1

09.4	M1: $K_a = \frac{[H^+][X^-]}{[HX]}$ or $[H^+] = \frac{K_a \times [HX]}{[X^-]}$ M2: amount of HX = 0.0500 mol M3: amount of HX after addⁿ of KOH = $0.05 - 3 \times 10^{-4} = 0.0497$ mol M4: amount of KX after addⁿ of KOH = $0.0136 + 3 \times 10^{-4} = 0.0139$ mol M5: $[H^+] = \left(\frac{1.41 \times 10^{-5} \times 0.0497}{0.0139} \right) = 5.04(15) \times 10^{-5}$ M6: $pH = -\log_{10} 5.04(15) \times 10^{-5} = 4.30$	M1: allow $[H^+] = \frac{K_a \times [\text{acid}]}{[\text{salt}]}$ M3: = $M2 - 3 \times 10^{-4}$ Answer to 2 decimal places If no attempt at M3 and M4 max 2 marks If M3 or M4 attempted using 3×10^{-4} max 4 (M1, M2, M3 or M4 and M6)	1 1 1 1 1 1
09.5	ratio $\frac{[HX]}{[X^-]}$ remains (almost) constant	Allow inverse expression	1

0	4
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Propanoic acid ($\text{C}_2\text{H}_5\text{COOH}$) is a weak acid.

The acid dissociation constant (K_a) for propanoic acid is $1.35 \times 10^{-5} \text{ mol dm}^{-3}$ at 25°C

0	4	.	1
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State the meaning of the term weak acid.

[1 mark]

0	4	.	2
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Give an expression for the acid dissociation constant for propanoic acid.

[1 mark]

K_a

0	4	.	3
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A student dilutes 25.0 cm^3 of $0.500 \text{ mol dm}^{-3}$ propanoic acid by adding water until the total volume is 100.0 cm^3

Calculate the pH of this diluted solution of propanoic acid.

Give your answer to 2 decimal places.

[4 marks]

pH _____



0	4	.	4
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A buffer solution with a pH of 4.50 is made by dissolving x g of sodium propanoate ($\text{C}_2\text{H}_5\text{COONa}$) in a solution of propanoic acid. The final volume of buffer solution is 500 cm^3 and the final concentration of the propanoic acid is $0.250 \text{ mol dm}^{-3}$

Calculate x in g

For propanoic acid, $K_a = 1.35 \times 10^{-5} \text{ mol dm}^{-3}$

[6 marks]

x _____ g

12

Turn over ►



Question	Answers	Additional comments/Guidelines	Mark
04.1	(Acid) partially or slightly ionises/dissociates (in water to form H ⁺ ions)	Allow – does not fully ionise/dissociate	1
04.2	(K _a) = $\frac{[\text{H}^+][\text{C}_2\text{H}_5\text{COO}^-]}{[\text{C}_2\text{H}_5\text{COOH}]}$	Allow [H ₃ O ⁺] for [H ⁺] Do not allow ()	1

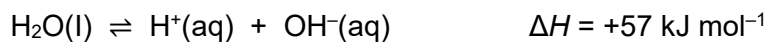
04.3	M1 [C ₂ H ₅ COOH] = <u>0.125</u> (mol dm ⁻³)	Allow consequential marking from wrong M1	1
	M2 [H ⁺] = $\sqrt{K_a \times [\text{C}_2\text{H}_5\text{COOH}]}$ OR [H ⁺] = $\sqrt{1.35 \times 10^{-5} \times 0.125}$	If [C ₂ H ₅ COOH] = 0.0125 (mol dm ⁻³) lose M1, allow M2, M3 = 4.108 x 10 ⁻⁴ and M4 = 3.39	1
	M3 [H ⁺] = 1.30 x 10 ⁻³ (mol dm ⁻³)		1
	M4 pH = $-\log_{10} (1.30 \times 10^{-3}) = 2.89$	Allow M4 = $-\log_{10} \text{M3}$ Answer must be to 2 decimal places	1

04.4	M1 $[H^+] = 10^{-4.5} = \underline{3.16 \times 10^{-5}} \text{ (mol dm}^{-3}\text{)}$		1
	M2 $[C_2H_5COO^-] = \frac{[C_2H_5COOH] K_a}{[H^+]}$	M2: Rearrangement	1
	OR $[C_2H_5COO^-] = \frac{0.250 \times 1.35 \times 10^{-5}}{3.16 \times 10^{-5}}$	If rearrangement incorrect, could score M1 M4 M5 and M6	
	M3 $[C_2H_5COO^-] = 0.1068 \text{ (mol dm}^{-3}\text{)}$		1
	M4 M_r sodium propanoate = 96		1
	M5 $n \text{ (sodium propanoate)} = 0.1068 \times 0.5 = 0.0534$		1
	M6 $\text{mass (sodium propanoate, x)} = 0.0534 \times 96 = 5.13 \text{ (g)}$	M6 Allow 5.09 to 5.14 (g)	1
	Alternative M5 and M6		
	M5 $0.1068 \times 96 = 10.25 \text{ (g)}$	M5 = M3 x M4	
	M6 $\text{Mass (sodium propanoate, x)} = \frac{10.25}{2} = 5.13 \text{ (g)}$	10.25 g scores 4 marks	
		If Henderson Hasselbach used M1 $pK_a = -\log(1.35 \times 10^{-5}) = 4.87$ M2 $\log[A^-] = pH - pK_a + \log[HA]$ M3 $\log[A^-] = 4.50 - 4.87 + (-0.60206) = -0.9717$ M4 $[CH_3CH_2COO^-] = 10^{-0.9717} = 0.1067$ M5 $M_r = 96(.0)$ M6 $x = 96.0 \times 0.1067/2 = 5.12 \text{ g}$	

0 6

This question is about pH.

Pure water dissociates slightly.



The equilibrium constant, $K_c = \frac{[\text{H}^+][\text{OH}^-]}{[\text{H}_2\text{O}]}$

The ionic product of water, $K_w = [\text{H}^+][\text{OH}^-]$

0 6

. 1

Explain why $[\text{H}_2\text{O}]$ is not shown in the K_w expression.

[1 mark]

Table 5 shows how K_w varies with temperature.

Table 5

Temperature / °C	$K_w / \text{mol}^2 \text{ dm}^{-6}$
10	2.93×10^{-15}
20	6.81×10^{-15}
25	1.00×10^{-14}
30	1.47×10^{-14}
50	5.48×10^{-14}

0 6

. 2

Explain why the value of K_w increases as the temperature increases.

[2 marks]



0	6	.	3
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Give the expression for pH.

Calculate the pH of pure water at 50 °C
Give your answer to 2 decimal places.

Explain why water is neutral at 50 °C

[4 marks]

Expression _____

Calculation

pH _____

Explanation _____

Question 6 continues on the next page

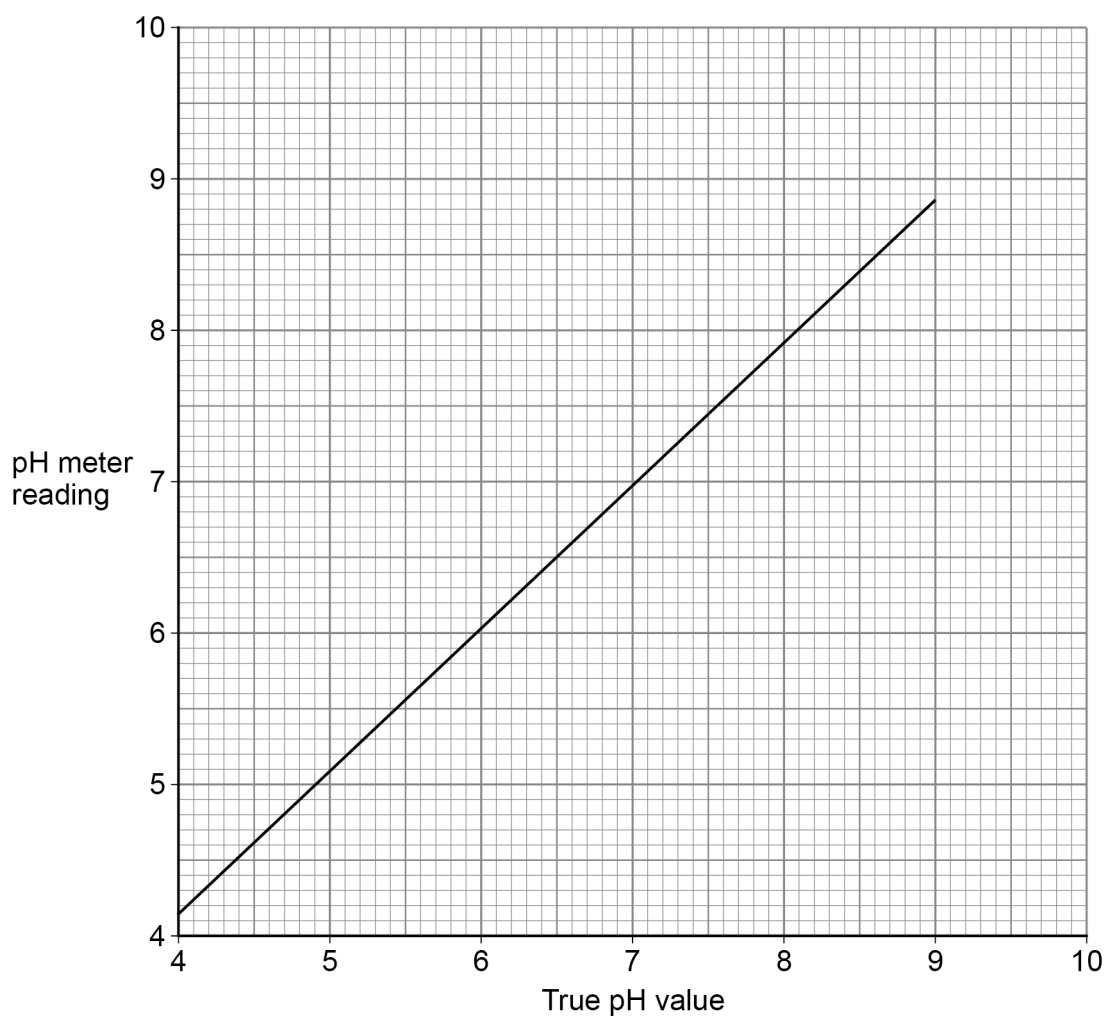
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A pH meter is calibrated using a calibration graph.
To create the calibration, the pH meter is used to measure the pH of separate solutions, each with a known, accurate pH.

Figure 3 shows the calibration graph.

Figure 3



0 6 . 4 Use **Figure 3** to give the true pH value when the pH meter reading is 5.6

[1 mark]

0 6 . 5 Suggest why the pH probe is washed with distilled water between each of the calibration measurements.

[1 mark]



0 6 . 6

The calibrated pH meter is used to monitor the pH during a titration of hydrochloric acid with sodium hydroxide.

Explain why the volume of sodium hydroxide solution added between each pH measurement is smaller as the end point of the titration is approached.

[1 mark]

Figure 4 shows the pH curve for a titration of hydrochloric acid with sodium hydroxide solution.

Figure 4

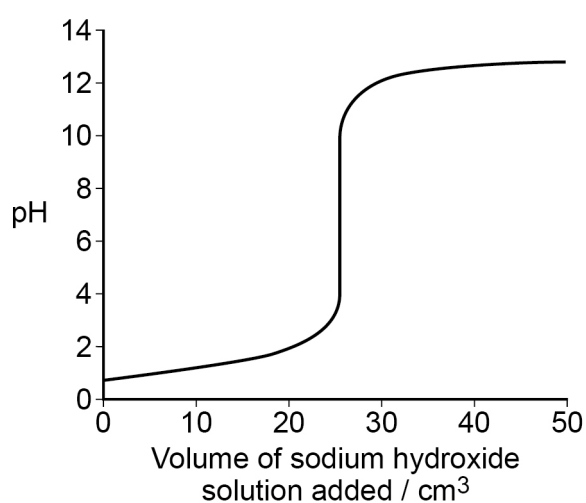


Table 6 shows data about some indicators.

Table 6

Indicator	pH range	Colour at low pH	Colour at high pH
Bromocresol green	3.8 – 5.4	yellow	blue
Phenol red	6.8 – 8.4	yellow	red
Thymolphthalein	9.3 – 10.5	colourless	blue

The student plans to do the titration again using one of the indicators in **Table 6** to determine the end point.

0 6 . 7

State why all three of the indicators in **Table 6** are suitable for this titration.

[1 mark]

Turn over ►



0	6	.	8
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36.25 cm³ of 0.200 mol dm⁻³ sodium hydroxide solution are added to 25.00 cm³ of 0.150 mol dm⁻³ hydrochloric acid.

Calculate the pH of the final solution at 25°C

$K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$ at 25°C

[5 marks]

pH _____

16



Question	Answers	Additional comments/Guidelines	Mark
06.1	[H ₂ O] is (almost) constant	Allow [H₂O] is (very) large in comparison (to [H⁺] and [OH⁻]) or [H₂O] is incorporated in K_w or $K_w = K_c[H_2O]$ or the equilibrium lies <u>very</u> much to the left. Ignore water has negligible dissociation Ignore [H₂O] = 1 or [H₂O] is very small	1

Question	Answers	Additional comments/Guidelines	Mark
06.2	M1 Equilibrium is endothermic (in forward direction)	Ignore more H ⁺ and OH ⁻ formed	1
	M2 <u>Equilibrium</u> shifts to the RHS to minimise/oppose temperature increase		1

Question	Answers	Additional comments/Guidelines	Mark
06.3	M1 pH = $-\log_{10}[H^+]$	M1 Allow pH = $-\log[H^+]$	1
	M2 $[H^+] = \sqrt{5.48 \times 10^{-14}} (= 2.34 \times 10^{-7})$	M2 $[H^+]^2 = 5.48 \times 10^{-14}$	1
	M3 pH = $-\log_{10} 2.34 \times 10^{-7} = \underline{6.63}$	M3 pH = $-\log_{10} M2$	1
	M4 $[H^+] = [OH^-]$ or Dissociation of each water molecule gives one H ⁺ and one OH ⁻	M4 Allow equal amounts of H ⁺ and OH ⁻	1

Question	Answers	Additional comments/Guidelines	Mark
06.4	5.55	Allow 5.5 to 5.6	1

Question	Answers	Additional comments/Guidelines	Mark
06.5	Different solutions must not contaminate each other or To wash off any residual solution/substance (which could interfere with the reading)	pH of previous solution doesn't contaminate new solution Ignore to make neutral/neutralise Ignore so as not to affect concentrations	1

Question	Answers	Additional comments/Guidelines	Mark
06.6	To avoid missing the end point Or (Very little pH change per cm ³ added at start) large change in pH (near end point)		1

Question	Answers	Additional comments/Guidelines	Mark
06.7	All have a colour change/pH range within the <u>steep/vertical</u> part of the titration curve	Colour change/pH range between pH 3 and 11	1

Question	Answers	Additional comments/Guidelines	Mark
06.8	M1 Amount of $\text{OH}^- = 36.25 \times 0.200 \div 1000 = 7.25 \times 10^{-3} \text{ mol}$ and Amount of $\text{H}^+ = 25.0 \times 0.150 \div 1000 = 3.75 \times 10^{-3} \text{ mol}$		1
	M2 Amount of excess $\text{OH}^- = 7.25 \times 10^{-3} - 3.75 \times 10^{-3}$ $= 3.50 \times 10^{-3} \text{ mol}$		1
	M3 $[\text{OH}^-] = (3.50 \times 10^{-3}) \div (61.25 \times 10^{-3}) (= 5.71 \times 10^{-2} \text{ mol})$	M3 $[\text{OH}^-] = (\text{M2}) \div (61.25 \times 10^{-3})$	1
	M4 $[\text{H}^+] = \underline{1.00 \times 10^{-14}} \div 5.71 \times 10^{-2} = 1.75 \times 10^{-13}$	M4 $[\text{H}^+] = 1.00 \times 10^{-14} \div \text{M3}$	1
	M5 pH = 12.76	M5 Allow pH = 12.8 M5 pH = $-\log_{10}(\text{M4})$ Alternative Method M4 p OH = 1.24 M5 pH = $14 - 1.24 = 12.76$	1

0	4
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This question is about acids and bases.

0	4	.	1
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Calculate the pH of a $0.150 \text{ mol dm}^{-3}$ solution of ethanoic acid at 25°C
Give your answer to 2 decimal places.

For ethanoic acid, $K_a = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$ at 25°C

[3 marks]

pH _____

0	4	.	2
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Strontium is an element in Group 2.

Calculate the pH of a $0.0100 \text{ mol dm}^{-3}$ solution of strontium hydroxide at 10°C
You may assume that strontium hydroxide is completely dissociated in this solution.

At 10°C the ionic product of water, $K_w = 2.93 \times 10^{-15} \text{ mol}^2 \text{ dm}^{-6}$

[3 marks]

pH _____



0 4 . 3

The pH of a barium hydroxide solution is lower at 50 °C than at 10 °C

At 50 °C a 25 cm³ sample of this barium hydroxide solution was neutralised by 22.45 cm³ of hydrochloric acid added from a burette.

Deduce the volume of this hydrochloric acid that should be added from a burette to neutralise another 25 cm³ sample of this barium hydroxide solution at 10 °C

[2 marks]

Circle (○) the correct answer.

> 22.45 cm³= 22.45 cm³< 22.45 cm³

Explain your answer

0 4 . 4

State how a buffer solution can be made from solutions of potassium hydroxide and ethanoic acid.

Give an equation for the reaction between potassium hydroxide and ethanoic acid.

State how this buffer solution resists changes in pH when a small amount of acid is added.

[3 marks]

How buffer solution is made

Equation

How buffer solution resists pH change

Turn over ►

0	4	.	5
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A buffer solution is made by adding 2.00 g of sodium hydroxide to 500 cm³ of 1.00 mol dm⁻³ ethanoic acid solution.

Calculate the pH of this buffer solution at 25 °C

Give your answer to 2 decimal places.

For ethanoic acid, $K_a = 1.74 \times 10^{-5}$ mol dm⁻³ at 25 °C

[5 marks]

pH _____

16



Question	Answers	Additional Comments/Guidelines	Mark
04.1	M1 $K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]} = \frac{[H^+]^2}{[CH_3COOH]}$		1
	M2 $[H^+] \left(= \sqrt{1.74 \times 10^{-5} \times 0.150} \right) = 1.62 \times 10^{-3} \text{ (mol dm}^{-3}\text{)}$		1
	M3 pH = 2.79	M3 = – log M2 answer must be to 2 decimal places	1 AO2

Question	Answers	Additional Comments/Guidelines	Mark
04.2	M1 $[OH^-] = \underline{2} \times 0.01 = 0.02$		1
	M2 $[H^+] \left(= \frac{K_w}{[OH^-]} = \frac{2.93 \times 10^{-15}}{0.02} \right) = 1.47 \times 10^{-13}$	allow $\frac{2.93 \times 10^{-15}}{M1}$	1
	M3 pH = 12.83	allow 12.8 M3 = – log M2 if pH = 12.5(3) allow 2 marks (not used factor of 2 in M1)	1 AO2

Question	Answers	Additional Comments/Guidelines	Mark
04.3	22.45 same $[\text{OH}^-]$ or same amount/number of OH^- ions		1 1 AO3

Question	Answers	Additional Comments/Guidelines	Mark
04.4	add excess ethanoic acid to KOH or add enough KOH to the ethanoic acid so that the acid is partially neutralised or add enough KOH so that the acid contains a mixture of ethanoic acid and ethanoate ions $\text{KOH} + \text{CH}_3\text{COOH} \rightarrow \text{CH}_3\text{COOK} + \text{H}_2\text{O}$ CH_3COO^- (from salt) reacts with (added) acid/ H^+	 allow $\text{KOH} + \text{CH}_3\text{COOH} \rightarrow \text{CH}_3\text{COO}^- \text{K}^+ + \text{H}_2\text{O}$ ignore equilibrium shifts	1 1 1 AO3

Question	Answers	Additional Comments/Guidelines	Mark
04.5	M1 (at start) $n(\text{NaOH}) = \frac{2.00}{40} = \underline{0.05}$ (mol) and $n(\text{CH}_3\text{COOH}) = \frac{500 \times 1.0}{1000} = \underline{0.5}$ (mol)		1
	M2 (after adding NaOH) $n(\text{CH}_3\text{COOH}) = (0.50 - 0.05) = 0.45$ (mol)		1
	M3 $n(\text{CH}_3\text{COO}^-) = (n \text{ NaOH}) = 0.05$ (mol)	M3 = $n(\text{NaOH})$ from M1	1
	$\text{M4 } [\text{H}^+] = \left(\frac{K_a \times [\text{CH}_3\text{COOH}]}{[\text{CH}_3\text{COO}^-]} \text{ or } \frac{1.74 \times 10^{-5} \times 0.9}{0.1} \text{ or } \frac{1.74 \times 10^{-5} \times 0.45/V}{0.05/V} \right)$ $= 1.57 \times 10^{-4} \text{ (mol dm}^{-3}\text{)}$	$\text{M4} = \frac{1.74 \times 10^{-5} \times \text{M2}}{\text{M3}}$ V cancels out so not needed in this expression	1
	M5 pH = 3.80	answer to 2 decimal places M5 allow 3.81 allow pH = $-\log \text{M4}$	1 AO2

04.5	Henderson–Hasselbach method M1 (at start) $n(\text{NaOH}) = \frac{2.00}{40} = \underline{0.05}$ (mol) and $n(\text{CH}_3\text{COOH}) = \frac{500 \times 1.0}{1000} = \underline{0.5}$ (mol) M2 (after adding NaOH) $n(\text{CH}_3\text{COOH}) = (0.50 - 0.05) = 0.45$ (mol) or $[\text{CH}_3\text{COOH}] = 0.9(0)$ M3 $n(\text{CH}_3\text{COO}^-) = n(\text{NaOH}) = 0.05$ (mol) or $[\text{CH}_3\text{COO}^-] = 0.1(0)$ M4 $\text{pH} = 4.759 + \log \frac{[\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]}$ or $4.759 + \log \frac{0.1}{0.9}$ or $4.759 + \log \frac{0.05/V}{0.45/V}$ M5 $\text{pH} = 3.80$	M3 = $n(\text{NaOH})$ from M1 V cancels out so not needed in this expression answer to 2 decimal places M5 allow 3.81	1 1 1 1 1 AO2
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Quick Quiz Instruction Guide

The data below are for your reference on which file you might want for your reference.

Please be noted that this is NOT the final paper and questions might be inaccurate

QQ Number	Topic	Paper
1	Kinetics	OCR 2020
2	Enthalpy	AQA 2020
3	Entropy	AQA 2020
4	Equilibria	AQA 2019,2020
5	Benzene	AQA 2021 CIE 2022
6	Alcohol	CCEA 2022
7	Phenol	Edexcel iAL 2023 October CIE 2021
8	Carbonyls	AQA 2019,2020
9	Acid Base Equilibria	